# PHYS0525 Analog and Digital Electronics – Spring 2026

## Assignment 8

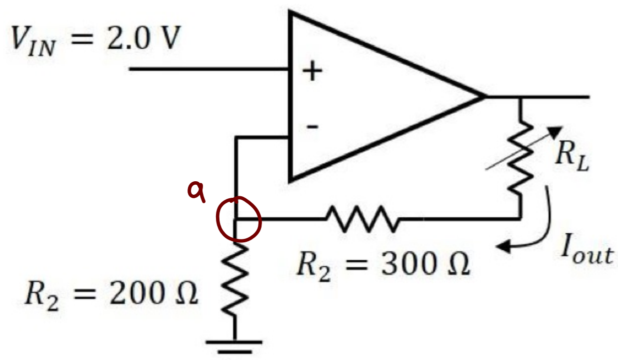
Due Monday, April 6, 2026

1) The circuit below can be thought of as a voltage controlled current source.

(a) Show that the current, $I_{out}$, through the load resistor, $R_L$, is independent of the value of $R_L$.

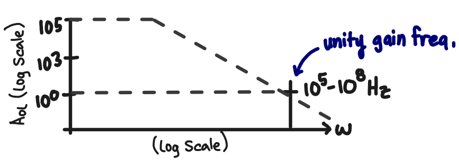
(b) For $V_{in} = 2$ Volts, what is the value of $I_{out}$?

(c) If the op amp has supply voltages $V_s$=10 V and $-V_s$= -10 V, what is the maximum current that can be supplied to a 100 Ω load resistor? What input voltage does this correspond to?




a.) **Properties of op-amps**

- Inputs are very high impedance ($10^6$ ~ $10^{14}$ Ω) (typically assume $Z_{in} \cong \infty$ [1-100 Ω]
- output impedance is low (1-100 Ω) (assume $Z_{out}$ is negligible)
- output $-V_s < V_{out} < +V_s$ (typically assume $A_{OL}$ is large $\rightarrow \infty$)




$V_{th} > 0$ ($V_{out} = +2V$)

$V_{th} < 0$ ($V_{out} = -2V$)

$V^- = V^+ = V_{in}$

∘ no current to inputs

@ a: $\frac{V_{in} - 0}{200\Omega} + \frac{V_{in} - V_L}{300\Omega} = 0$

↳ $\frac{3V_{in}}{2} + (V_{in} - V_L) = 3V_{in} + 2V_{in} - 2V_L = 0$

↳ $V_L = \frac{5}{2}V_{in}$

and $I_{out} = I @ R_L = \frac{V_L - V_{in}}{300\Omega} \Rightarrow \frac{\frac{5}{2}V_{in} - V_{in}}{300}$

so, $I_{out} = \frac{\frac{3}{2}V_{in}}{300} \Rightarrow \frac{V_{in}}{200}$ (independent of L)

b.) For $V_{in} = 2V$ : $I_{out} = \frac{2}{200} = \frac{1}{100}A = 0.01 A = 10mA$

c.) $V_{out} = V_L + I_{out} \cdot R_L$

↳ $V_{out} = \frac{5}{2}V_{in} + \frac{V_{in}}{200} \cdot 100$

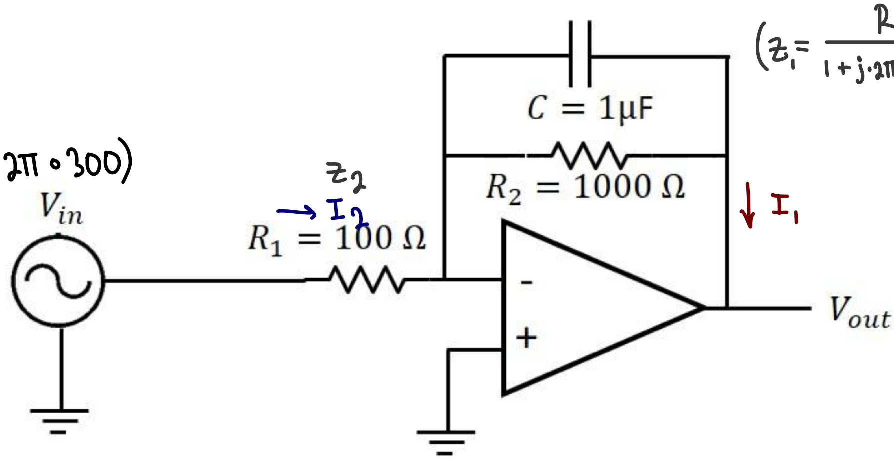
$V_{out} = \frac{5}{2}V_{in} + \frac{1}{2}V_{in} = 3V_{in}$

and $V_{in} = \pm 10V$, so: $V_{out} \simeq \frac{10}{3}V = 3.33V$

2) The circuit below has an AC input voltage with an amplitude of 1V at a frequency $\omega = 2\pi \times 300\ \mathrm{s}^{-1}$. What is the amplitude of the output voltage? You may assume the op-amp is ideal. (Hint: You can consider this circuit as an inverting amplifier with the gain given by the ratio of the feedback impedance to the input impedance.)

$\left(Z_1 = \dfrac{R}{1 + j \cdot 2\pi \cdot 300 \cdot 3\,k\Omega \cdot 1\mu F}\right)$

(1V @ $2\pi \circ 300$)




Golden Rule

$V_{in}^+ = V_{in}^- = 0 \qquad I_1 = I_2$

$I_2 = \dfrac{V_{in} - V_{in}^-}{R_2} = \dfrac{V_{in}}{R_2} = I_1$

$V_{out} = -I_1 R_1 = -\dfrac{V_{in}}{R_2} \cdot R_1$

Gain

$V_{out} = \left(-\dfrac{R_1}{R_2}\right) \cdot V_{in} \simeq \dfrac{|Z_1|}{|Z_2|} \circ V_{in} \Rightarrow$

Inverting Amplifier

$$\frac{\left\|\left[\dfrac{1}{1 + \left(j\frac{2\pi}{300} 1000 \cdot 1E\text{-}6F\right)}\right]\right\|}{|100\,\Omega|} \cdot 1V \Rightarrow \frac{\dfrac{1}{\sqrt{1 + \left(\frac{2\pi}{300} \cdot 1000 \cdot 1E\text{-}6\right)^2}}}{\sqrt{100^2}} = 4.686 \circ 1V \simeq \underline{4.07V}$$